October University for Modern Sciences and Arts (MSA)
Department of Electronics and Communication
Course: ECE552 Mobile Communication Systems
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Problem Set #2 Solution
Principles of Mobile Communications

Question #2.1

Suppose that a mobile station is moving along a straight line between two base stations BS1 and BS2, as shown in Fig. 2.5. The distance between the base stations is D = 10 Km. Assume P_o = 0 dB_m, d_o = 1 m and n = 2. Assume the minimum usable signal level for acceptable voice quality at the base station receiver is P_{rmin} = -74 dBm, mobile average speed is 100 km/h and the threshold level used by the switch for handoff initiation is P_{rHO} = -73 dBm. do the following:

i) Determine the minimum required margin, ΔP to assure that calls are not lost due to weak signal condition during handoff.
ii) Find the distance which at handoff occurs and the maximum time, Δt required by MS for processing handoff.
iii) Find the maximum time, Δt required by MS for processing handoff, if the user moves with speed 120 km/h.

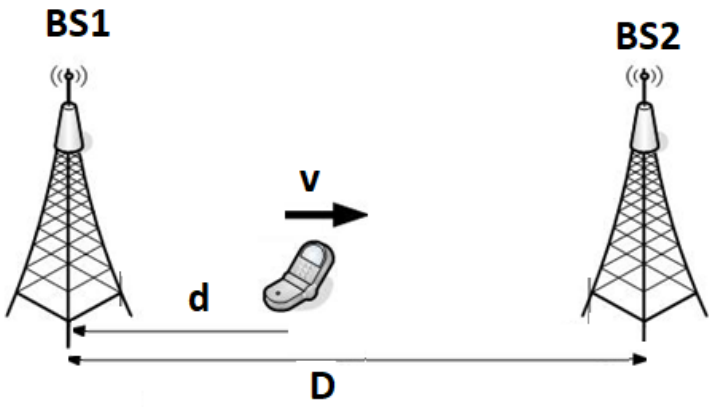


Fig. 2.5

Solution

a) The minimum required margin to assure that calls are not lost due to weak signal condition during handoff.

$$\Delta P = P_{rHO} - P_{rmin} = -73 + 74 = 1 \ dBm$$

b) The distance which at handoff start is calculated as:

$$P_r(d_{HO}) = P_r(d_o) - 10 * n \log_{10}(d_{HO}/d_o) \quad \Rightarrow -73 = 0 - 10 * 2 * \log_{10}(d_{HO}/1) \ \ so \quad d_{HO} = 4.46\,\text{km}$$

The distance which at handoff occurs is calculated as:

$$P_r(d_{\max}) = P_r(d_o) - 10 * n \log_{10}(d_{\max}/d_o) \quad \Rightarrow -74 = 0 - 10 * 2 * \log_{10}(d_{\max}/1) \quad so\ d_{\max} = 5.012\,\text{km}$$

$$\Delta d \ = d_{\max} - d_{H.O} = 545 \ meters$$

The maximum time, Δt required by MS for processing handoff is:

$$\Delta t = \frac{d_{nax} - d_{HO}}{v_{MS}} = \frac{545}{27.78} = 19.62 \ \text{ sec}$$

c) The maximum time, Δt required by MS for processing handoff, if the user moves with speed 120 km/h.

Since the MS is moving by (120 km/h) = 33.3 m/s, so

$$\Delta t = \frac{d_{nax} - d_{HO}}{v_{MS}} = \frac{545}{33.3} = 16.37 \ \text{ sec}$$

So $\Delta P = P_{rHO} - P_{rmin}$ can be increased

***Question #2.2*[HW]**

Suppose that a mobile station is moving along a straight line between two base stations BS1 and BS2 as in Fig. 2.5. The distance between the base stations is D = 2500 m. Assume P_o = 5 dBm, do = 10 m and n = 2.5. Assume the minimum usable signal level for acceptable voice quality at the base station receiver is P_{rmin} = -90 dBm, and the threshold level used by the switch for handoff initiation is (time required to complete a handoff, once that received signal level reaches the handoff threshold P_{rHO} is Δt = 4.0 seconds).

a) Determine the min distance for acceptable signal level.
b) Determine the minimum required margin $\Delta = P_{rHO} - P_{rmin}$ to assure that calls are not lost due to weak signal condition during handoff. Assume that the base stations antenna heights are negligible compared to the distance between the mobile and the base station.
c) Describe the effect of the margin $\Delta = P_{rHO} - P_{rmin}$ on the performance of cellular systems.

Question #2.3

Assume each user generates 0.2 Erlangs of traffic, How many users can be supported for blocking probability, P_r = GOS = 0.2 % for the following number of trunked channels, N_{chcell} in a blocked calls cleared system? Refer to Table 3.1.

a) $N_{chcell} = 5$ b) $N_{chcell} = 10$ c) $N_{chcell} = 20$ d) $N_{chcell} = 100$.

Table 3.1 Capacity of an Erlang B system

Number of Channels N_{chcell}	Capacity, A (Erlangs) for GOS = 0.01	= 0.005	= 0.002	= 0.001
2	0.153	0.105	0.065	0.046
4	0.869	0.701	0.535	0.439
5	1.36	1.13	0.900	0.762
10	4.46	3.96	3.43	3.09
20	12.0	11.1	10.1	9.41
24	15.3	14.2	13.0	12.2
40	29.0	27.3	25.7	24.5
70	56.1	53.7	51.0	49.2
100	84.1	80.9	77.4	75.2

Solution

From Table 3.1, we can find the total capacity in Erlangs for the 0.2% GoS for different numbers of channels. By using the relation, $A = N_{uscell}\, A_u$, we can obtain the total number of users that can be supported in the system.

a) Given $N_{chcell} = 5$, $A_u = 0.2$, GoS = 0.002. So, we obtain A = 0.9.
Therefore, total number of users, $N_{uscell} = A / A_u = 0.9 / 0.2 = 4$ user.

b) Given $N_{chcell} = 10$, $A_u = 0.2$, GoS = 0.002. So, we obtain A = 3.43.
Therefore, total number of users, $N_{uscell} = A / A_u = 3.43 / 0.2 = 16$ user.

c) Given $N_{chcell} = 20$, $A_u = 0.2$, GoS = 0.002. So, we obtain A = 10.1
Therefore, total number of users, $N_{uscell} = A / A_u = 10.1/0.2 = 50$ user.

d) Given $N_{chcell} = 100$, $A_u = 0.2$, GoS = 0.002. So, we obtain A = 77.4
Therefore, total number of users, $N_{uscell} = A / A_u = 77.4/0.2 = 387$ user.

Question #2.4 **[HW]**

Assume each user generates 0.5 Erlangs of traffic, How many users can be supported for blocking probability, P_r = GOS = 1 % for the following number of trunked channels, N_{chcell} in a blocked calls cleared system? Refer to Table 3.1.

a) $N_{chcell} = 5$ b) $N_{chcell} = 10$ c) $N_{chcell} = 20$ d) $N_{chcell} = 100$.

Solution

As done in Question #3.7.

Question #2.5

An urban area has a population of one million residents. Three competing trunked mobile networks (systems A, B, and C) provide cellular service in this area. System A has 350 cells (N_{cell}) with $N_{chcell} = 24$ channels each, system B has 90 cells (N_{cell}) with $N_{chcell} = 40$ channels each, and system C has 35 cells (N_{cell}), each with N_{chcell} =70 channels. Find the number of users that can be supported at 0.5 % blocking if each user averages two calls per hour at an average call duration of three minutes. Assuming that all three trunked systems are operated at maximum capacity, compute:

a) The total number of subscribes for each system
b) The percentage market penetration of each cellular provider

Solution Each user makes 2 calls/hour × 3 minutes = 6 minutes/hour = 0.1 Erlangs
Probability of blocking, P_b = GoS = 0.5 % = 0.005

a) For system A:

Number of channels per cell used in the system, N_{chcell} = 24 Channel

For GoS = 0.005 and N_{chcell} = 24, from the Erlang B Table:

The total carried traffic, A = 14.2 Erlangs.

Therefore, the number of users that can be supported per cell, N_{uscell} is:

$N_{uscell} = A / A_u = 14.2/0.1 = 142$ user.

Since there are 350 cells, the total number of subscribers that can be supported by system A is equal to 142 × 350= 39700 subscribers.

For system B:

Number of channels per cell used in the system, N_{ch} = 40 Channel

Traffic intensity per user, $A_u = \lambda H = 2 \times (3/60) = 0.1$ Erlangs

For GoS = 0.005 and N_{chcell} = 40 Channel, from the Erlang B chart, Table:

The total carried traffic, A = 27.3 Erlangs.

Therefore, the number of users that can be supported per cell, N_{uscell} is:

$N_{uscell} = A/ A_u = 27.3/0.1 = 273$ user

Since there are 90 cells, the total number of subscribers that can be supported by system B is equal to 273× 35 = 24,570 subscribers.

For system C:

Number of channels per cell used in the system, N_{chcell}= 70 Channel

Traffic intensity per user, $A_u = \lambda H = 2 \times (3/60) = 0.1$ Erlang.

For GOS = 0.01 and N_{chcell} = 70, from the Erlang B chart, Table:

The total carried traffic, A = 53.7 Erlang.

Therefore, the number of users that can be supported per cell, N_{uscell} is:

$N_{uscell} = A/ A_u = 53.7/0.1 = 537$ subscribers

Since there are 35 cells, the total number of subscribers that can be supported by system C is equal to 537 × 35 = 18,795 subscribers.

b) Therefore, total number of cellular subscribers that can be supported by these three systems are:

Total subscribers = 39700+ 24,570 + 18,795 = 83,065 user or subscriber

- Since there are one million residents in the given urban area and the total number of cellular subscribers in system A is equal to 47280 therefore:

  System A of market penetration = 39700/ 1,000,000 = 3.97 %
- Similarly, market penetration of system B is:

  System B of market penetration = 24,570 / 1,000,000 = 2.46 %
- The market penetration of system C is:

  System C of market penetration = 18,795 / 1,000,000 = 1.88 %
- The market penetration of the three systems combined is given by:

  The three systems market penetration = 83,065 / 1,000,000 = 8.3 %

Question #2.6 **[HW]**

An urban area has a population of one million residents. Three competing trunked mobile networks (systems A, B, and C) provide cellular service in this area. System A has 300 cells (N_{cell}) with N_{chcell} = 40 channels each, system B has 80 cells (N_{cell}) with N_{chcell} = 70 channels each, and system C has 40 cells (N_{cell}), each with N_{chcell} =100 channels. Find the number of users that can be supported at 0.2 % blocking if each user averages two calls per hour at an average call duration of three minutes. Assuming that all three trunked systems are operated at maximum capacity, compute the percentage market penetration of each cellular provider.

Solution As done in Question #3.9.

Question #2.7

A certain city has an area of 1,500 square kilometers and is covered by a cellular system using a 12-cell reuse pattern. Each cell has a radius of 5 km and the city is allocated 25 MHz of spectrum for uplink with a channel bandwidth of 200 kHz and the same for downlink. Assume a GOS of 0.2 % for an Erlang B system is specified. If the offered traffic per user, A_u = 0.5 Erlangs, compute the following:

a) The number of clusters in the service area.
b) The number of channels per cell.
c) Traffic intensity of each cell.
d) The maximum carried traffic.
e) The total number of users that can be served for 1% GOS
f) The number of mobiles per unique channel (where it is understood that channels are reused).

Solution

a) For the system, Given:

Total coverage area, $Area_{city}$ = 1500 km^2 m, and cell radius, R = 5 km.
The cell area (hexagon) can be shown to be $2.5981R^2$, thus:
$Area_{cell} = 2.5981 \times (5)^2 = 64.95$ km^2
Cluster area $Area_{cluster}$ = N x Cell area = 12 x 64.95 = 779.43 km^2
Hence, the total number of clusters are N_{cls}= 1500/779.43 = 1.92 ≈ 2 cluster

b) # of channels per cluster, $N_{chcls} = N_{chnet}$ = 25,000,000/200,000 = 125 channel
of channels per cell, N_{chcell}= 125/ 7 = 17.86 ≈17 channel

c) Given: N_{chcell} = 17, and GoS = 0.002

From the Erlang B Table, we have traffic intensity per cell, A_{cell} is:

A_{cell} = 10.1 Erlang/cell

d) Maximum carried traffic = traffic intensity per cell x 7 x number of clusters

A_{total} = 10.1 x 7 x 2 = 141.14 Erlang.

e) The traffic per user, A_u = 0.5 Erlangs

of users per cell (Cell Capacity), $N_{uscell} = A_{cell} / A_u = 10.1 / 0.5 = 20$ user

Total number of users, N_{usnet} = Total traffic / traffic per user:

$N_{usnet} = A_{total} / A_u = 1141.14 / 0.5 = 282$ user

or $N_{usnet} = N_{uscell} \times N \times N_{cls} = 20 \times 7 \times 2 = 282$ user.

f) Number of mobiles per channel = number of users/number of channels

= 282 / 17 = 16.59 ≈ 16 mobiles/channel.

Question #2.8 **[HW]**

A certain city has an area of 1,500 square kilometers and is covered by a cellular system using a 12-cell reuse pattern. Each cell has a radius of 5 km and the city is allocated 75 MHz of spectrum for uplink with a channel bandwidth of 200 kHz and the same for downlink. Assume a GOS of 0.5 % for an Erlang B system is specified. If the offered traffic per user, A_u = 0.4 Erlangs, compute the following:

a) The number of cells in the service area and the number of channels per cell.
b) The traffic intensity of each cell and the maximum carried traffic.
c) The total number of users that can be served for 0.5 % GOS
d) The number of mobiles per unique channel (where it is understood that channels are reused).

Solution As done in Question #3.11.